Ex. Every natural number is either $\underbrace{\text{even}}_{\exists k \in \mathbb{N} \text{ s.t } n = 2k}$ or $\underbrace{\text{odd}}_{\exists k \in \mathbb{N} \text{ s.t } n = 2k+1}$.

$(n = 2k-1)$

Pf. ① By induction

Use $p(n) =$ $n$ is even or odd.

($P(n)$ is true $\forall n$).

Base Case

$P(1) = 1$ is either even or odd.

Since $1 = 2 \cdot 1 - 1$

this is a true statement.

Induction Step

$$\underset{\text{true}}{P(n-1)} \Longrightarrow \underset{\text{true}}{P(n)}$$

Since $P(n-1)$ is true,

$n-1$ is either even or odd.

- If even: $n-1 = 2k$ ($k \in \mathbb{N}$)

  then $n = 2k+1$

  $\Rightarrow$ $n$ is odd, so $P(n)$ is true.

- If odd: $n-1 = 2k-1$

  then $n = 2k$ $\Rightarrow$ $n$ is even, so $P(n)$ is true.

By "induction axiom"
every $n$ is either even or odd.

$\blacksquare$

② By WOP.
let $A = \{n \in \mathbb{N} : n \text{ is neither even nor odd}\}$
If $A = \phi$, then all $n \in \mathbb{N}$ is even or odd. So we're done.

Now assume $A \neq \phi$. Then we can apply WOP, and let $m$ be the smallest element of $A$.

We know that $m \neq 1$, b/c 1 is odd $\Rightarrow m > 1$
Then look at $m-1$ $(< m)$
By the minimality of $m$ in $A$
we know $m-1 \notin A$
i.e. $m-1$ is either even or odd.

- $m-1$ even: $m-1 = 2k$ for some $k$
$\Rightarrow m = 2k+1$, i.e. $m$ odd, $m \notin A$
- $m-1$ odd: $m-1 = 2k-1$ for some $k$
$\Rightarrow m = 2k$, i.e. $m$ even, $m \notin A$.

Contradictions, So $m \notin A \Rightarrow A = \phi$
$\Rightarrow$ Every $n \in \mathbb{N}$ is even or odd.

Back to Strong Induction: Give $P(n)$.

if ① $P(1)$ true.

② $P(1) \dots P(n-1)$ true $\Longrightarrow P(n)$ true.

Then, $P(n)$ true $\forall n \in \mathbb{N}$.

Ex. ① A class of $n \geq 12$ students can be split up into groups of 4 or 5.

Pf. Strong Induction.

Base Case

$n = 12$: $12 = 4 + 4 + 4$

Inductive step

Suppose $12, 13, \dots, n-1$ students can be spilt up into groups of 4 or 5.

Assume $P(11+n)$, $P(12+n)$, $P(13+n)$, $P(14+n)$

$\Longrightarrow P(15+n)$ is true (from $P(11+n)$).

By strong induction, $P(n)$ true $\forall n \geq 12$.

② Show that every natural number $n \geq 1$ can be written as a product of primes.

Pf. We'll use strong induction

Base Case $n = 2$.

2 is prime, which is the product of one prime.

Inductive Step

Suppose that all numbers $2, 3, \ldots, n-1$ can be written as a product of primes.

Now look at $n$.

If $n$ is prime, than we're done.

If $n$ is not prime, then $\exists\, a, b \in \mathbb{N}$, s.t. $n = a \cdot b$.

Since $n$ is not prime, $a, b \neq 1, n$.

By strong induction,

$$\left.\begin{aligned} a &= p_1 p_2 \cdots p_s \\ b &= q_1 q_2 \cdots q_t \end{aligned}\right\} \text{Product of primes}$$

$n = ab = p_1 \cdots p_s q_1 \cdots q_t$

Thm. Induction $\Longleftrightarrow$ Strong induction.

Pf. Assume that induction is true

Suppose that we're given

① $P(1)$ is true.

② $\underbrace{P(1), \cdots, P(n-1)}_{\text{true}} \Rightarrow \underbrace{P(n)}_{\text{true}}$

Let $Q(n)$ be a new statement.

$P(1), \cdots, P(n)$ are all true.

(If $Q(n)$ true $\Longrightarrow$ $P(n)$ true)

We will show that $Q$ is true $\forall n \in \mathbb{N}$

by induction ($P$ is true $\forall n \in \mathbb{N}$.)

Base Case. $Q(1)$ is true $\Longleftrightarrow$ $P(1)$ is true.

· Inductive Step:

Need to verify $\underbrace{Q(n-1)}_{\text{true}} \Longleftrightarrow \underbrace{Q(n)}_{\text{true}}$

$\Rightarrow$ $Q(n-1)$ true.

$\Longleftrightarrow$ $P(1), \cdots, P(n-1)$ true.

$\xRightarrow{\text{ass.}}$ $P(n)$ is true

$Q(n)$ true $\Longrightarrow$ $P$ true.

($\Leftarrow$) Assume Strong induction

If $P(1)$ is true,

$P(1), \ldots, P(n-1)$ true $\Rightarrow P(n)$ true.

To show induction

Suppose $P(1)$ true

$P(n-1)$ true $\Rightarrow P(n)$ true

$\underbrace{P(1), \ldots, P(n-1)}_{\text{true.}} \Rightarrow P(n)$ true.